

Jitter Protocol — Provisional Patent Research Dossier

Bottom line: No existing patent or product combines keystroke biometrics with content provenance.

Across 14 relevant prior art patents, dozens of competitors, and hundreds of market participants, no one uses typing dynamics to certify that a human authored specific text content. The closest prior art (BehavioSec US10,068,088B2) detects bots via behavioral biometrics but for fraud prevention, not content provenance. The market opportunity is validated by BioCatch's **\$1.3B acquisition**, (Forge) (Sacra) GPTZero's **253% ARR growth**, (Sacra) and the structural collapse of CAPTCHAs — AI now solves them at **96–99.8% accuracy** (Checkmarx) versus humans at 50–86%. (CyberPeace Foundation) (CoinDesk) Regulatory tailwinds are strong: the EU AI Act Article 50 mandates machine-readable AI content labeling by **August 2, 2026**, (Pearl Cohen) (European Commission) and C2PA's deliberate exclusion of human identity from its core spec (C2pa) creates the exact gap Jitter Protocol fills.

SECTION 1 — PATENT PRIOR ART

14 relevant patents identified; none combine all six Jitter Protocol differentiators

Extensive searching across Google Patents, USPTO, and patent databases surfaced **14 directly relevant prior art patents**. Every patent found focuses on identity authentication ("who is typing?") or session-level fraud detection — **none address content provenance** ("a human typed this specific content"). The six Jitter Protocol differentiators — content provenance purpose, embeddable widget, accumulating passport, reveal-don't-block, privacy-first on-device processing, and binary human/bot output — appear in no existing patent individually or in combination.

Highest-relevance prior art

BehavioSec — US10,068,088B2 (filed 2012, granted 2018, now LexisNexis Risk Solutions). This is the **closest prior art** and warrants the most careful claim differentiation. The patent covers differentiating humans from bots using behavioral biometrics including dwell time, trajectory, pressure, acceleration, velocity, and "jitter data." It uses fuzzy evaluation to detect human-like versus machine-like input behavior. (google) **Critical differences from Jitter Protocol:** BehavioSec's patent is for banking fraud detection, not content provenance; it sends data to a centralized "behaviometric server" (not on-device/privacy-first); it has no embeddable widget concept, no accumulating passport, no reveal-don't-block approach, and no content-level provenance output. However, the overlap in signal types (including the word "jitter") requires precise claim drafting.

Adobe — US10,503,919B2 (filed 2017, granted 2019). Uses keystroke biometrics as a second authentication factor for electronic signatures. (google) Explicitly references TypingDNA as a service provider. (Google Patents) **Differs because:** it verifies identity (who signed), not content provenance; it's tied to e-signature workflows, not general content creation; no widget architecture, passport, or binary human/bot output.

Human vs. Bot Gesture Fingerprinting — US10,908,929B2 (filed 2019, granted 2021). Captures sensor data to make a binary human/bot determination ([Google Patents](#)) — the closest patent to Jitter Protocol's output format. **Differs because:** it targets ad fraud counting, uses gesture-based (not keystroke-specific) analysis, has no content provenance focus, no embeddable widget, no passport accumulation.

Moderate-relevance prior art

Akamai — US20210037048A1 (filed 2019). Uses JavaScript client-side collection of keystrokes, mouse movements, and gyroscope data with deep learning for bot/human classification. Server-side processing, web traffic focus, not content provenance.

SRI/DSO — US8,332,932B2 (filed 2007, granted 2012). Foundational patent on dwell time, flight time, and "dwell tendency" (rate of change). ([Google Patents](#)) Pure identity authentication. ([Google Patents](#)) Jitter Protocol uses similar measurements but for fundamentally different purpose.

BehavioSec — US9,301,140B1 (filed 2013, granted 2016). Defines the "behavioral server" architecture with continuous profile updates. Centralized processing model versus Jitter's on-device approach.

Cognitive Fingerprints — US20150169854A1 (filed 2013, DARPA-funded). Extracts cognitive typing rhythm features from keystroke dynamics. ([Google Patents](#)) **Directly relevant to Jitter Protocol's "cognitive pauses" feature** — but used for personal identification, not human/bot distinction.

BehavioSec — US20200074053A1 (filed 2019). Continuous authentication from touch/soft keyboards at OS kernel level. ([Google Patents](#)) Mobile-focused, identity verification purpose.

Correlating Cognitive Biometrics — US20160006730A1 (filed 2014). Multi-modal continuous verification including forensic linguistic analysis. **Captures WHAT you type** (content) for analysis — directly contrary to Jitter Protocol's privacy-first principle of capturing only HOW you type.

Free-Text Behavioral Authentication — US20220245225A1 (filed 2022). Free-text keystroke dynamics authentication using hold time and flight time. ([Google Patents](#)) Identity authentication, not content provenance. However, the free-text approach is relevant since Jitter Protocol similarly must work with arbitrary typed content.

EP2477136A1/B1 (filed 2011). Continuous identity verification via keystroke dynamics using di-graph flight times. ([Google Patents](#)) Standard identity verification post-login.

CyberArk — US9,342,687B2 (granted 2016). Detects synthetic (non-physical-keyboard) keystrokes by comparing OS-level physical input records against application-received keystrokes. ([Google Patents](#)) Different mechanism entirely — detects injected input rather than analyzing typing patterns.

NeuraMetrix — US20170116399A1 (filed 2015). Typing cadence authentication storing timing data in RAM. **Notable overlap:** captures HOW you type, not WHAT — same privacy approach as Jitter Protocol. But purpose is identity authentication, not content provenance.

Onenigma — WO2018029523A1 (filed 2016). Touch-coordinate-based user identification on touch interfaces. [Google Patents](#) **Interesting overlap:** progressive enrollment from first use parallels Jitter Protocol's accumulating passport concept.

TypingDNA and BioCatch patent portfolios

TypingDNA holds **3 registered patents** in computing/calculating. [Crunchbase](#) A February 2024 patent grant for 2FA/MFA via typing biometrics was announced, [TypingDNA](#) but the specific patent number could not be confirmed. ⚠️ The full claims of this patent should be obtained and reviewed. All TypingDNA technology is identity authentication, not content provenance. [Infosecurity Magazine](#)

BioCatch holds **92+ registered patents** focused on behavioral biometrics for financial fraud detection.

[PR Newswire](#) Patents cover mouse movements, typing patterns, touch gestures, and cognitive behavior analysis — all oriented toward fraud prevention in banking, not content provenance. [Biocatch](#) [Crunchbase](#)

Novelty assessment and claim drafting guidance

Not found in any prior art: (1) Using keystroke dynamics specifically to certify human authorship of content rather than identify a person; (2) an embeddable web component (`<jitter-input>`) for keystroke biometric content verification; (3) cross-session confidence accumulation ("passport"); (4) tagging bot content visibly rather than blocking it; (5) combining privacy-first on-device processing with content provenance; (6) bridging keystroke biometrics with C2PA content credentials for text.

Areas requiring careful claim drafting: BehavioSec's US10,068,088B2 (human/bot detection via overlapping signal types), US10,908,929B2 (binary human/bot output), Adobe's US10,503,919B2 (keystroke biometrics in document workflows), and the cognitive fingerprints patent US20150169854A1 (cognitive rhythm analysis). Jitter Protocol's strongest claims should emphasize the **combination** of all six differentiators — no existing patent combines even three.

⚠️ **Flag:** TypingDNA's February 2024 patent grant number should be obtained and its full claims reviewed before filing.

SECTION 2 — CURRENT COMPETITIVE LANDSCAPE (2025–2026)

TypingDNA: active but small, purely identity-focused

TypingDNA remains operational [TypingDNA](#) with **11–50 employees** in Brooklyn, NY, [Crunchbase](#) selling keystroke biometric authentication products: **ActiveLock** (continuous endpoint authentication), **Verify 2FA** (typing-based two-factor), [Typingdna](#) and an **Authentication API**. [Infosecurity Magazine](#) Total funding is **\$8.6M** (Series A), with no new rounds reported. [Crunchbase](#) In February 2025 they launched ActiveLock 3.5 "Fortress" with dual-layer security. [IntechOpen](#) [Typingdna](#) In September 2025 they were named a Sample Vendor in the **Gartner Hype Cycle for Digital Identity 2025** for "Passive Behavioral Biometrics." [Typingdna](#) [TypingDNA](#) Their CEO, Raul Popa, stated passive behavioral analytics is "on the productivity plateau." [TypingDNA](#) **They**

have zero products or apparent interest in content provenance — their entire focus is identity authentication.

BioCatch: thriving at \$1.3B valuation, banking fraud only

BioCatch was acquired by **Permira for \$1.3B** in May 2024 (The Times of Israel) (Sacra) and exceeded **\$185M ARR** in Q4 2025 — their best quarter ever. They added **90 new customers in 2025** including Wells Fargo, bringing total customers to 340+ financial institutions. Three of four largest US banks now use BioCatch. (Biocatch) (Sacra) The platform analyzes **3,000+ data points** per session (Sacra) across 17 billion user sessions monthly, protecting 660M+ people. (Biocatch) They launched **BioCatch Trust Argentina** (first inter-bank behavioral fraud intelligence network in the Western Hemisphere) and **Scams360** for detecting social engineering. (Biocatch) BioCatch is entirely focused on banking fraud — not content provenance, not general web bot detection.

BehavioSec: absorbed into LexisNexis

BehavioSec was **acquired by LexisNexis Risk Solutions on May 3, 2022**. (PR Newswire) Their technology is now embedded in **LexisNexis ThreatMetrix** (PR Newswire) for fraud prevention within a \$10B+ revenue organization (RELX, \$75B market cap). (Forgepoint Capital) No independent product development for content provenance.

New entrants (2023–2026)

Roundtable (founded 2023, Y Combinator-backed) is the most directly relevant new entrant. Founded by two Princeton PhDs, (Y Combinator) it offers "**Proof of Human**" — invisible bot detection using behavioral biometrics (Roundtable) including keystroke dynamics. Claims **87% bot detection accuracy** versus reCAPTCHA's 69% and Cloudflare Turnstile's 33% in head-to-head testing. (Roundtable) Pricing starts at \$99/month with one-line API integration. (Roundtable) **Key difference from Jitter Protocol:** Roundtable does session-level bot detection, not per-content provenance attestation.

Self raised a **\$9M seed** (November 2025) for zero-knowledge proof-of-humanity using biometric passports.

(FinTech Global) **Humanity Protocol** uses blockchain and ZK proofs for Sybil resistance. Both focus on identity verification, not content authorship.

Bot detection is failing on major platforms

The scale of the problem is staggering. An estimated **30% of online reviews globally are fake**, (Shapo) (Capital One Shopping) with AI-generated reviews growing **80% month-over-month** since June 2023. (Shapo) Fake reviews are projected to cost consumers **\$787 billion in 2025**. (Shapo) (Capital One Shopping) Amazon blocked 200M+ suspected fake reviews in 2022, investing \$500M+ and hiring 8,000 employees (Roast a Profile) (Capital One Shopping) — yet **61% of electronics** (Thrive Agency) and **64% of supplement reviews** were still flagged as unreliable in 2025 analyses. (Roast a Profile) Google blocked **170M policy-violating reviews** in 2023 (45% YoY increase). (Shapo) Yelp detected ~500K AI-generated reviews in 2025, (Martech Cube) a 58% increase in Unusual Activity Alerts year-over-year. (Martech Cube) Reddit lacks any centralized bot detection system, relying on community moderators and third-party tools while LLM-generated "mention activity" floods subreddits. (Pangram Labs)

CAPTCHAs are structurally broken

AI bots now solve CAPTCHAs with 96–99.8% accuracy while humans achieve only 50–86%.

(CyberPeace Foundation) (Jetir) ETH Zurich researchers demonstrated **100% success rate** against reCAPTCHA v2 image challenges. (CoinDesk) Commercial bypass services (CapSolver, CaptchaAI) (Kasada) charge as little as **\$0.0007 per 1K solves** at 99.5%+ success rates. (Captchaai) Cloudflare Turnstile achieved only **33% bot detection accuracy** in independent testing. (Roundtable) (Roundtable) The fundamental problem: reCAPTCHA v3's behavioral analysis is defeated by AI that generates statistically indistinguishable interaction patterns. (BroadChannel) (Checkmarx) Google reportedly moved reCAPTCHA management to Google Cloud in 2025 and slashed the free tier from 1M to 10K assessments/month. (Roundtable)

Content provenance for text: the gap Jitter Protocol fills

No company currently uses keystroke biometrics for text content provenance. Adjacent solutions include:

AI text detectors (GPTZero, Originality.ai) analyze statistical properties of output text. GPTZero claims ~99% accuracy (GPTZero) with 8M+ users and an estimated **\$24M ARR** growing 253% YoY. (Sakra) Originality.ai claims 96–99% accuracy. (Originality.AI) **Both are fundamentally different from Jitter Protocol:** they analyze the output, not the creation process, and are inherently reactive — evasion improves as AI models improve. (arXiv)

Worldcoin/World (\$240M+ funded, 25M users) uses iris-scanning Orb devices to create World ID — proof that an account belongs to a verified human. (Blockchain Council +2) **But World ID cannot prove a verified human authored specific content.** A verified human could still post AI-generated text.

C2PA/Content Credentials (200+ members) provides cryptographic content provenance (C2pa) — but designed for media files, not typed text, and explicitly cannot verify human authorship (see Section 3).

⚠ Critical academic challenge to note

A January 2026 paper by David Condrey (arXiv:2601.17280) — **"On the Insecurity of Keystroke-Based AI Authorship Detection"** — directly challenges the approach Jitter Protocol proposes. It demonstrates two attack classes: **(1) copy-type attacks** (human transcribes LLM text, producing authentic motor signals) and **(2) timing-forgery attacks** (bots sample inter-keystroke intervals from human distributions), achieving **≥99.8% evasion rates** against five classifiers. (arXiv) The paper's key finding: "These systems confirm a human operated the keyboard, but not whether that human originated the text." (arXiv) It recommends provenance systems bind writing process to semantic content through revision trajectories, challenge-response protocols, or cryptographic commitments. (arXiv) **The patent filing should explicitly address how Jitter Protocol defends against copy-type and timing-forgery attacks** — potentially through the edit behavior, error patterns, and cognitive pause signals that go beyond simple timing.

A countervailing paper by Deane (2026, Journal of Educational Measurement) found keystroke dynamics **can** distinguish bona fide versus transcribed/paraphrased responses, "significantly outperforming humans" at detecting LLM-assisted writing when examining cognitive demands and writing strategies. (ResearchGate)

SECTION 3 — C2PA & VIDEO PROVENANCE

Cameras shipping with C2PA content credentials

Leica pioneered C2PA implementation: the **M11-P** (C2PA) (October 2023, world's first), **M11-D** (September 2024), and **SL3-S** (January 2025) all ship with C2PA built in. (c2paviewer)

Sony has the broadest camera support: **Alpha 1**, **Alpha 1 II**, **Alpha 9 III** (C2PA) (firmware v2.00+), **Alpha 7 IV** and **Alpha 7S III** (firmware v3.00+), (c2paviewer) and the **PXW-Z300** (July 2025, world's first camcorder with C2PA video support). (Content Authenticity Initiative) Sony's implementation was initially limited to news agencies via a licensed service.

Nikon Z6 III received C2PA via firmware (Canon Rumors) v2.00 (August 2025) but was **suspended in September 2025** after a critical vulnerability: a security researcher demonstrated that the Multiple Exposure feature allowed signing AI-generated images and images from non-C2PA cameras. Nikon revoked all issued C2PA certificates. (PetaPixel) ⚠️ **Service has NOT been restored as of early 2026.** The Z9 and Z8 were announced but do not yet support C2PA.

Canon EOS R1 and R5 Mark II — C2PA announced in firmware v1.1.0 (July 2025). (Newsshooter) ⚠️ **Status uncertain:** Canon Europe's press release confirms C2PA support, (Canon) but Canon USA stated the inclusion "was not intended and would not be coming at this time." Canon Japan suspended the firmware for a video playback bug. May be available exclusively for registered news agencies. (Canon)

Google Pixel 10 (September 2025) is the **first smartphone with full C2PA signing by default** for every photo, hardware-backed via Titan M2 chip, achieving the highest C2PA Assurance Level 2. (Gadget Hacks)

Samsung Galaxy S25 (January 2025) applies C2PA credentials **only to AI-edited/generated photos** (Android Authority) (Generative Edit, Sketch to Image, Portrait Studio) — not all camera captures.

Apple iPhone: NO C2PA support whatsoever as of early 2026. Apple has not joined the C2PA coalition or announced Content Credentials. This is a critical gap given iPhones capture the majority of the world's photographs.

Fujifilm GFX100S II and **X-T50** (2024) support C2PA. (c2paviewer) (C2PA) Panasonic joined CAI in 2025 (Content Authenticity Initiative) but no shipping cameras yet.

C2PA adoption is early-stage — comparable to early HTTPS

The Content Authenticity Initiative community has grown to **6,000+ members** as of January 2026.

(contentauthenticity) (Content Authenticity Initiative) C2PA has **200+ formal members**. (Defense) Specification v2.2 was published May 2025; v2.3 is in development, and the spec is under review by ISO for standardization. (LOC) Cloudflare became the first major CDN to implement C2PA, covering **~20% of the web**.

(Content Authenticity Initiative) AI platforms implementing C2PA include OpenAI (DALL-E), Google (Gemini/Imagen), Adobe Firefly, and Stability AI. (c2paviewer) **Midjourney does NOT implement C2PA.**

(C2PA Viewer) LinkedIn displays Content Credentials in its feed. (c2paviewer) However, most social media platforms still strip metadata on upload ("Am I Cited") — the **#1 adoption obstacle**. Engineer Tim Bray noted: "I had a freaking hard time finding [a photo with content credentials]. There are very few Content Credentials out there on the Internet." (ongoing by Tim Bray)

Adobe Content Credentials in practice

Adobe has embedded Content Credentials across its product line: **Photoshop** (opt-in per document, records editing actions), **Lightroom** (exported images), **Firefly** (automatically applied to 100% AI-generated images), (Contentauthenticity) **Premiere Pro**, and **Adobe Stock**. A free web app (**Adobe Content Authenticity**, public beta October 2024) allows batch-applying credentials, inspecting images, and includes a Chrome extension. (Adobe Newsroom) "Durable Content Credentials" use invisible watermarks plus cloud storage for recovery even when metadata is stripped. (C2pa) (Adobe Newsroom) An enterprise version launched in 2025. Open-source tools include **c2pa-rs** (Rust SDK), (C2PA) **C2PA JavaScript SDK**, and **c2patool** (CLI).

The human-behind-the-camera gap — where Jitter Protocol fits

C2PA explicitly and by design does NOT prove a human was behind the camera or keyboard. This is a deliberate architectural decision for privacy. In C2PA v2.0 (January 2024), **all references to human/organization identity were removed from the core specification**. The C2PA FAQ states: "The core C2PA Content Credentials specification does not support attribution of content to individuals or organizations, so that it can remain maximally privacy-preserving." (C2pa) The C2PA Security Considerations document states: "The identity of a signatory is not necessarily a human actor." (C2pa)

The Nikon Z6 III exploit perfectly demonstrates this gap: a security researcher successfully got a C2PA-enabled camera to cryptographically sign an AI-generated image of a pug flying a plane, passing all verification checks. (PetaPixel) (Digital Camera World) C2PA proves a **device** created content — not that a **human** initiated the capture.

CAWG (Creator Assertions Working Group) was established in February 2024 as a separate body to handle identity assertions as optional extensions, using W3C Verifiable Credentials and Decentralized Identifiers. But identity remains controversial within the community due to privacy concerns for vulnerable creators (whistleblowers, human rights defenders).

This gap is precisely what Jitter Protocol addresses for text content. C2PA is device-centric trust; Jitter Protocol would be human-centric biometric trust. These are complementary: C2PA handles the device/tool chain, Jitter Protocol handles the human-provenance layer.

C2PA coalition membership and governance

C2PA was co-founded (Content Credentials) in February 2021 by Adobe, Arm, BBC, Intel, Microsoft, and Truepic. (C2pa +2) Current steering committee includes Sony (March 2022), Publicis Groupe (June 2023), Google (February 2024), OpenAI (May 2024), (Business Wire) Meta (September 2024), and Amazon (September 2024). (PR Newswire) (C2pa) The coalition operates as a Joint Development Foundation project (C2pa) (C2pa) under the Linux Foundation with royalty-free, open specifications. (Content Authenticity Initiative) The Content Authenticity

Initiative (CAI), founded by Adobe in 2019, serves as the community/implementation arm (C2pa +2) — CAI develops tools, C2PA writes the standard.

SECTION 4 — HARDWARE BIOMETRICS & WEB APIs

Hall Effect keyboard analog data is accessible via WebHID — proven working

WebHID API browser support: Chrome/Edge ✓ (since v89, March 2021), Opera ✓, Firefox ✗ (actively resisted), Safari ✗ (no plans), Mobile browsers ✗. (Testmu) (caniuse) Global coverage is ~39% (Chromium desktop only). WebHID is a WICG spec, not a W3C standard. (WICG) (caniuse)

The critical finding for the patent: Hall Effect keyboards CAN expose per-key analog pressure data (0–255 range, representing 0–100% actuation depth) to web browsers via WebHID. This has been demonstrated with multiple working implementations: **wooting-js** (TypeScript library reading analog input from Wooting keyboards), (Socket) **kb-hall** (reads per-key analog values from MCHOSE magnetic keyboards, renders real-time heatmap), (GitHub) **WebHMK/Libhmk** (open-source HE keyboard library with browser configurator), (Notebookcheck) and **Macrolev** (open-source analog keyboard with web configurator). The technical mechanism: HE keyboards expose analog data through **vendor-specific HID usage pages** (e.g., 0xFF00), bypassing Chrome's block on protected usages (standard keyboard usage page 0x01). This requires HTTPS, a user gesture to trigger device selection, and explicit user approval via browser picker dialog. (GitHub)

Gamepad API exposes analog trigger pressure with ~96% browser coverage

The Web Gamepad API has ~95.8% **global browser coverage** (Chrome, Edge, Firefox, Safari including iOS). (caniuse) For **PS5 DualSense controllers**, trigger pressure (L2/R2) is exposed as (buttons[6].value) and (buttons[7].value) with a **0.0–1.0 analog range** (MDN Web Docs) — web apps can read how hard triggers are pressed. (MDN Web Docs) Full analog stick data is accessible. However, DualSense adaptive trigger resistance is output-only (cannot be read), advanced haptics are not accessible via the standard API, (Hardware Test Pro) and gyroscope/accelerometer data requires WebHID (Chromium only). **Xbox controllers** similarly expose analog trigger pressure at 0.0–1.0 via the Gamepad API. (GamePad Tester Pro) No permission dialog is required — controllers activate after first user interaction. (MDN Web Docs) (GamePad Tester Pro)

Mobile sensors: accelerometer/gyroscope available, touch pressure is dead

Accelerometer and gyroscope data are accessible through DeviceMotionEvent and DeviceOrientationEvent (web.dev) with ~95%+ **browser coverage**. iOS Safari requires explicit (DeviceMotionEvent.requestPermission()) since iOS 13 (must be called from a user gesture). (Trys Mudford) (OpenReplay) Android Chrome currently auto-grants access but plans to eventually require permission. (Google Groups) HTTPS is required on all browsers. The Generic Sensor API (Accelerometer, Gyroscope classes) offers a cleaner interface but is Chrome/Edge only. (GitHub)

Touch pressure is effectively dead on all current mobile platforms. Apple removed 3D Touch hardware starting with iPhone 11 (September 2019) (Wikipedia) — the last iPhones with pressure sensing were the XS

and XS Max. [MacRumors](#) Haptic Touch is purely time-based and exposes no pressure levels. [MacRumors](#) On Android, virtually no current phones include display-level pressure sensors; the [Touch.force](#) API exists but returns 0 or a constant on modern devices. [jvaspring](#) The only remaining Apple device with web-accessible pressure data is **Apple Pencil on iPad** via [Touch.force](#) in Safari.

[Touch.radiusX](#)/[radiusY](#) on Android can serve as a rough proxy for pressure (larger contact area correlates with harder press), but this lacks precision for biometric applications. [jvaspring](#)

Summary: viable hardware expansion paths

Data Source	API	Browser Coverage	Viability
HE keyboard analog (per-key 0–255)	WebHID	~39% (Chromium desktop)	✅ Proven working
Controller trigger pressure (0.0–1.0)	Gamepad API	~96%	✅ Excellent
Mobile accelerometer/gyroscope	DeviceMotion/Orientation	~95%+	✅ Good (permission required on iOS)
Mobile touch pressure (iOS)	Touch.force	N/A	❌ Hardware removed since 2019
Mobile touch pressure (Android)	Touch.force	API exists	❌ No current hardware support

⚠️ **Flags:** WebHID's future in Firefox/Safari is uncertain — Mozilla has shown no implementation intent. [Mozilla Connect](#) Chrome may tighten sensor permissions to match Safari's model. [Google Groups](#) Hall Effect keyboard analog access depends on manufacturer choice to expose data via vendor-specific HID usage pages.

SECTION 5 — REGULATORY TAILWINDS

EU AI Act Article 50: AI content labeling mandatory by August 2, 2026

The EU AI Act (Regulation 2024/1689) is the world's most comprehensive AI regulation. **Article 50** is the central provision relevant to Jitter Protocol:

- **Art. 50(2):** Providers of generative AI must ensure outputs (images, audio, video, **text**) are **marked in a machine-readable format and detectable as artificially generated**. Solutions must be "effective, interoperable, robust, and reliable." [European Commission](#)

- **Art. 50(4):** Deployers must disclose when content is AI-generated, including **AI-generated text published to inform the public on matters of public interest**.
- **Penalties for non-compliance:** Up to **€15 million or 3% of worldwide annual turnover**. (Europa)

The phased timeline: prohibited AI practices applied February 2025; governance and GPAI model obligations August 2025; **transparency obligations (Art. 50) become enforceable August 2, 2026**; (Pearl Cohen) high-risk AI product rules August 2027. The European Commission published a first Draft Code of Practice on AI content labeling on December 17, 2025, promoting a **multilayered approach** combining visible disclosures with invisible machine-readable techniques. (European Commission) Final Code expected June 2026.

⚠ The EU Digital Omnibus proposal (November 2025) could delay some provisions to 2027. (IAPP)

US federal: regulatory vacuum, state laws filling the gap

No comprehensive federal AI disclosure or labeling law exists. Biden's AI Executive Order 14110 (October 2023) was **revoked by Trump on January 20, 2025**. (Wikipedia) Trump EO 14179 (January 2025) emphasized removing "barriers to innovation." (Federal Register) Trump EO 14365 (December 2025) actively seeks to **preempt state AI laws** through a federal framework and directed DOJ to establish an AI Litigation Task Force to challenge state laws. (Paul Hastings LLP) (Kslaw) However, an executive order cannot independently preempt state law — Congressional action or court decisions are required. (Paul Hastings LLP)

The only significant federal AI law enacted is the **TAKE IT DOWN Act** (May 2025), criminalizing nonconsensual AI-generated intimate imagery. Multiple AI labeling bills have been introduced but none enacted: AI Labeling Act (U.S. Senator Brian Schatz) (S.2691), COPIED Act, (Future of Privacy Forum) REAL Act, (FedScoop) AI Disclosure Act. (Holistic AI)

California leads state-level AI regulation

California Bot Disclosure Law (SB 1001) — in effect since July 1, 2019. (Perkins Coie) Makes it unlawful to use a bot to communicate with a person in California "with the intent to mislead about its artificial identity" for commercial transactions or election influence. (CA) Requires "clear, conspicuous" disclosure. (Salar Atrizadeh) Applies to platforms with 10M+ monthly US visitors. (LegiScan) Penalties up to **\$2,500 per violation**. (Financier Worldwide) ⚠ **No known enforcement actions** despite being in effect for nearly 7 years.

California AI Transparency Act (SB 942/CAITA) — enacted, effective date **delayed to August 2, 2026** (aligned with EU AI Act) by AB 853. (Privacy + Cyber + AI) Requires generative AI providers with 1M+ monthly CA users to offer free AI detection tools, (Plura) (Jones Day) mandatory latent watermarks (provider name, version, timestamp, unique ID), and visible labeling options. Penalties **\$5,000 per violation per day**. (Gadgetsgigabytesandgoodwill) ⚠ **Original law does not cover pure text content.**

Other notable state laws: **Colorado AI Act** (effective February/June 2026, high-risk AI systems), **Utah AI Policy Act** (effective May 2024, first US state to enact major AI disclosure), (Stackcyber) **New York RAISE Act** (passed legislature June 2025, returned with amendments by governor December 2025), **Texas RAIGA** (effective January 2026).

28 states have laws addressing deepfakes in political communications. (Stackcyber) Five states (California, Maine, New Hampshire, New York, Utah) have enacted chatbot-specific disclosure laws.

International regulatory landscape

South Korea enacted the **AI Framework Act** (effective January 22, 2026) — the first comprehensive AI legislation in Asia-Pacific. (Future of Privacy Forum) Requires operators to label products created by generative AI and provide clear notifications for deepfakes. Human- or machine-readable labeling accepted. (IAPP)

China has the most prescriptive requirements: AI Labeling Rules (effective September 1, 2025) require both explicit and implicit labels on AI-generated content. (IAPP) (Cookie-Script)

UK has **no specific AI content labeling legislation**. The Online Safety Act 2023 focuses on illegal content and child safety. Ofcom confirmed AI-generated content falls within scope when it constitutes user-generated content, (Pinsent Masons) and the UK government announced plans (February 2026) to close loopholes for AI chatbots. (365i Web Design)

Canada's AIDA died when Parliament was prorogued January 2025 following PM Trudeau's resignation. (SafeAI-Aus) No enacted federal AI legislation; reintroduction expected but no timeline. **Australia** has paused standalone AI legislation, relying on voluntary guidance and existing technology-neutral laws. (SafeAI-Aus)

Regulatory demand for proof-of-humanity tools

Multiple regulatory frameworks create **direct or indirect demand** for tools like Jitter Protocol:

- EU AI Act Article 50 requires content be "detectable as artificially generated" — logically implying tools that distinguish AI from human content
- The EU Draft Labelling Code explicitly states providers should "make available verification tools enabling users and third parties to assess the provenance of content"
- Bot disclosure laws (CA SB 1001, Utah SB 149) establish the legal premise that consumers have a right to know whether they're interacting with humans or machines
- Platform compliance obligations under the EU Digital Services Act and UK Online Safety Act require transparency measures

⚠️ **No existing legislation explicitly requires "proof-of-humanity" tools.** The demand is derivative — these laws require labeling of AI content, which logically requires tools to distinguish AI from human content.

Platform-level adoption accelerating

Major platforms are implementing AI content labeling: **Meta** auto-labels AI images on Facebook/Instagram when C2PA metadata is detected. **YouTube** requires creators to label synthetic content (enforcement began early 2025). **TikTok** was the first video platform to implement C2PA (May 2024), adding invisible watermarks in late 2025 and labeling **1.3 billion+ videos** to date. **Google** applies SynthID watermarks to AI-generated media.

SECTION 6 — MARKET SIZE

Behavioral biometrics: ~\$2.5–3.1B in 2025, growing 20–27% CAGR

Multiple analyst firms converge on a market size of approximately **\$2.5–3.1B in 2025**, projected to reach **\$7–18B by 2030–2033** depending on forecast horizon. Grand View Research estimates **\$9.92B by 2030** at 27.3% CAGR. Straits Research projects **\$18.46B by 2033** at 26% CAGR. Keystroke dynamics is specifically noted as the **fastest-growing sub-segment** at ~28% CAGR. North America holds ~33.6% market share; Asia-Pacific is the fastest-growing region at ~28–29% CAGR.

Bot detection/management: ~\$2.5–3.9B in 2025

Market estimates vary by segment definition. Broader "bot management" (including security, mitigation, and detection) is estimated at **\$2.5–3.9B in 2025**. Narrower "bot detection" is **\$1–1.9B in 2025**. Mordor Intelligence projects bot detection reaching **\$4.52B by 2030** at 26.2% CAGR. Key players include Imperva (Thales), Akamai, Cloudflare, HUMAN Security (~\$1.5B estimated valuation), and DataDome (\$82M total funding).

Content authenticity/provenance: emerging market, ~\$1.9B

This is a nascent but rapidly formalizing market. Dataintelo estimates the content authenticity verification market at **\$1.92B in 2024** growing at 18.7% CAGR. The Business Research Company published a formal C2PA market report in 2026. **Gartner listed digital provenance among its top 10 technology trends** for the next five years. ⚠️ Content provenance market sizing is very new; treat early estimates cautiously.

The problem is massive and accelerating

51% of all web traffic was automated bots in 2024 — the first time bots surpassed human traffic in a decade (Imperva Bad Bot Report). Bad bot traffic has risen for **six consecutive years**. An estimated **30% of online reviews are fake**, costing consumers **\$770.7B in 2025**. Deepfake incidents grew from 500K (2023) to **8M (2025)** — 900% annual growth. AI-generated phishing achieves a **54% click-through rate** versus 12% for traditional phishing (4.5× more effective). FBI IC3 reported **\$16.6B in cybercrime losses** in 2024 (33% YoY increase). Deloitte projects GenAI-enabled fraud growing from \$12.3B to **\$40B by 2027**.

Industry spending patterns

Education: Turnitin generates **~\$203M revenue** (2024), serving 16,000+ institutions and 71M students. California alone tracked \$15M+ in Turnitin spending across 57 institutions. Their AI detector has reviewed **200M+ papers** since April 2023.

Financial services: BioCatch at ~\$160M ARR serving 280+ institutions. Financial services is the most targeted industry for account takeover attacks (22% of all ATO incidents).

E-commerce: Amazon spends **\$500M+ annually** combating fake reviews with 8,000+ dedicated employees. Retail faces 59% bad bot traffic.

Advertising: Click fraud projected at **\$114B in 2025**, growing to \$172B by 2028. Bot fraud surged **101% YoY** in North America.

Recent funding rounds (2023–2026)

Company	Round	Amount	Date	Valuation
BioCatch	Acquisition (Permira)	—	May 2024	\$1.3B
HUMAN Security	Growth Equity II	\$50M	Oct 2024	~\$1.5B est.
DataDome	Series C	\$42M	Mar 2023	—
World (Worldcoin)	Multiple rounds	\$240M total	—	—
GPTZero	Series A	\$10M	Jun 2024	—
Self	Seed	\$9M	Nov 2025	—

GPTZero's trajectory is particularly relevant: estimated **\$24M ARR** in 2025, up 253% YoY from \$6.8M, profitable within 18 months with a 34-person team and 4M+ registered users. This validates strong market demand for AI content detection — and Jitter Protocol's process-level approach would complement rather than compete with output-level detection.

Combined TAM for Jitter Protocol

Jitter Protocol sits at the intersection of three converging markets with a combined adjacent TAM of **~\$7–10B in 2025, growing to \$20–40B+ by 2030–2033**. Validation signals include BioCatch's \$1.3B acquisition (behavioral biometrics in a single vertical), GPTZero's 253% ARR growth (AI content detection demand), Worldcoin's \$240M raise (proof-of-humanity), and C2PA's trajectory toward ISO standardization.

Conclusion

This research reveals a **distinct, defensible patent position** for Jitter Protocol. No existing patent combines keystroke biometrics with content provenance — the entire prior art corpus treats typing dynamics as an identity tool, not a content attestation tool. The competitive landscape confirms this gap: TypingDNA, BioCatch, and BehavioSec authenticate identity; GPTZero and Originality.ai analyze output text; C2PA verifies devices; Worldcoin verifies personhood. None prove that a human authored specific text content in real-time.

Three developments make the timing urgent. First, CAPTCHAs are structurally broken — AI solves them better than humans. Second, the EU AI Act's Article 50 transparency obligations take effect August 2, 2026, creating enforceable demand for machine-readable AI content detection. Third, C2PA deliberately excluded human

identity from its core spec in January 2024, confirming that the "human behind the keyboard" gap is acknowledged but unaddressed by the industry's leading provenance standard.

The patent filing should address the Condrey 2026 paper's copy-type attack challenge head-on, emphasizing that Jitter Protocol's analysis of edit behavior, error patterns, cognitive pauses, and revision trajectories goes beyond simple keystroke timing — binding the writing process to semantic content creation. The strongest claims should emphasize the **novel combination** of all six differentiators, as no existing patent combines even three. Hardware expansion via WebHID to Hall Effect keyboards is technically proven today, offering a future claims pathway to pressure-curve biometrics with per-key analog data (0–255 range) already accessible in Chromium browsers.